

BALU NIJAMPATNAM

Email: balunijampatnam09@gmail.com | Phone: +91-6302627030

LinkedIn: [linkedin.com/in/balu-nizampatnam](https://www.linkedin.com/in/balu-nizampatnam)

EDUCATION

Bachelor of Technology

Aug 2024 - May 2028

Indian Institute of Information Technology Design and manufacturing Jabalpur

Computer Science and Engineering

CPI - 8.8

TECHNICAL SKILLS

Languages: C++(proficient), Java(Basic), Python(Basic)

CoreCS: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Databases: SQL, MongoDB

Tools: Git, GitHub, VSCode

Relevant Coursework: Operating Systems, Computer Networks

PROJECTS

Real-Time Crowd Vision & Analytics System | Computer Vision Team Project

- Python, OpenCV, YOLOv8, NumPy, REST APIs
- Developed a real-time crowd monitoring system to analyze multiple camera feeds and generate crowd analytics for crowded environments.
- Designed a video processing pipeline to handle live streams and derive key metrics such as people count and crowd trend prediction using historical data.

Role: Crowd Detection & Density Estimation

- Implemented a person detection and counting module using YOLOv8 and OpenCV to detect individuals from live video streams.
- Estimated crowd density using bounding-box area information from detected individuals.
- Integrated the detection pipeline with REST APIs to send real-time crowd analytics data.

High-Performance Low-Latency Limit Order Book Engine

- C++, Algorithms, System Design
 - Developed a low-latency trading engine simulator that processes buy and sell orders and automatically executes trades when prices match, emulating stock exchange systems.
 - Implemented price-time priority matching logic to ensure fair and deterministic trade execution between buyers and sellers.
 - Designed a real-time order book data structure to track active bid and ask orders and update market state with minimal latency after each trade.
 - Validated the engine through simulations of partial fills, order cancellations, and price-crossing scenarios.
-

CODING PROFILES

LeetCode: leetcode.com/balu | 250+ problems solved | Rating: 1665(Max:1720)

CodeChef: [codechef.com/users/balu](https://www.codechef.com/users/balu) | Rating: 1411 (Max: 1451)

Codeforces: codeforces.com/profile/balu | Rating: 1011 (Max: 1102)

GitHub: github.com/balu

ACHIEVEMENTS

- Secured Global Rank 665 in CodeChef Starters 218 (Rated Contest)
 - Ranked 3761 / 42,306 participants in LeetCode Weekly Contest 492 (Top ~9%)
 - Maintained strong academic performance with Semester CPI 9.2 and Overall CPI 8.8 in . B.Tech CSE
-